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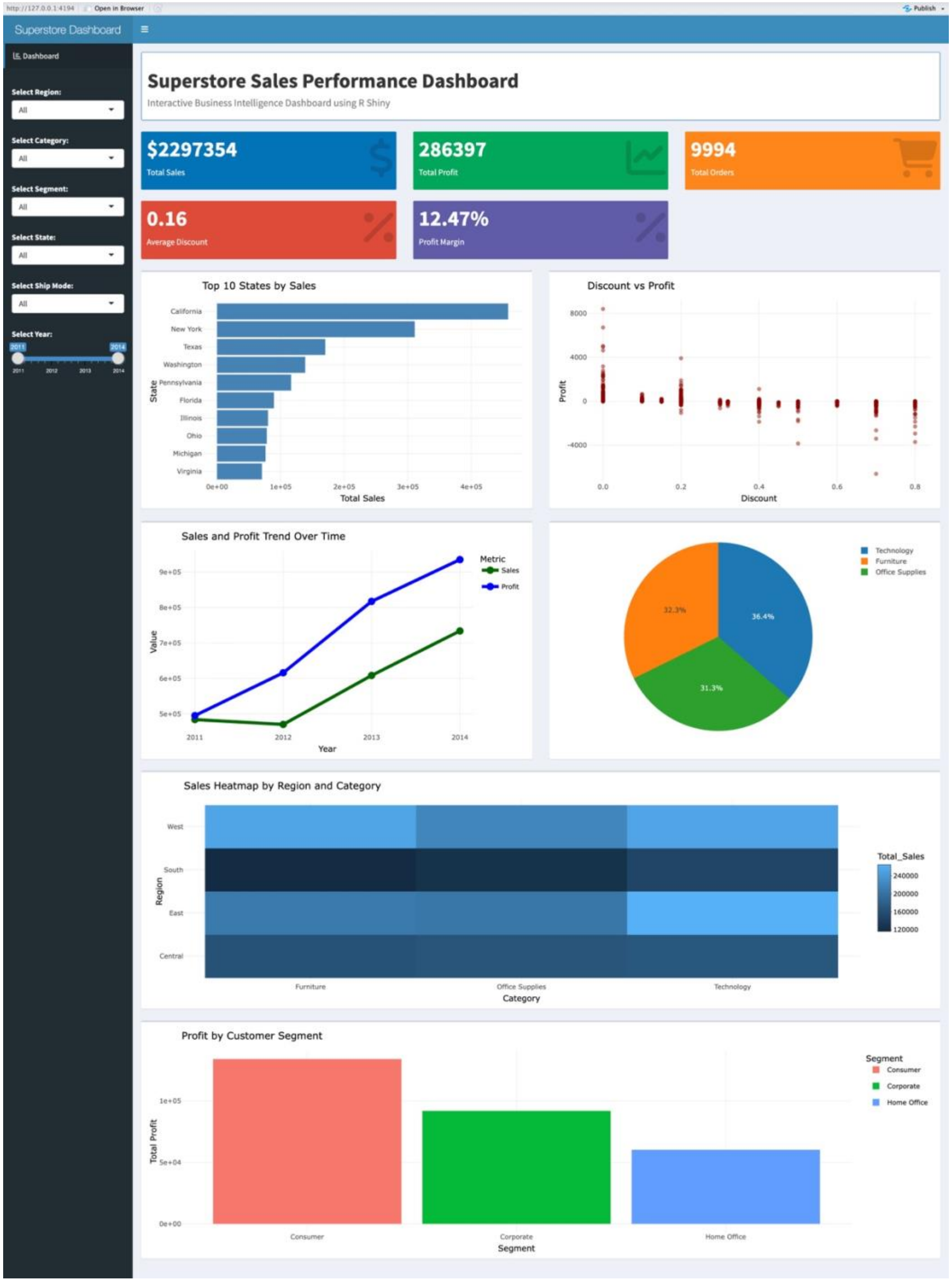
Page 1: Dashboard Screenshot



This dashboard was developed using R Shiny to analyse Superstore sales performance. Interactive filters for Region, Category, Segment, State, Ship Mode, and Year allow users to explore the data dynamically. Key insights are presented through KPI cards and multiple visualisations, including sales trends, discount analysis, regional performance, and customer segment analysis.

Page 2: Problem Design

Although Superstore generates strong sales across different regions and product categories, the main business problem is not simply whether the company can generate revenue, but whether revenue growth is translating into sustainable profitability. High sales performance may still be accompanied by weak profit margins due to excessive discounting, inefficient product categories or differences in customer profitability. This means that some regions, categories or customer segments may contribute strongly to sales while contributing relatively little to overall profit. Therefore, the key business challenge is identifying which parts of the business are truly driving profitable growth and which areas may be weakening long-term financial performance. The dashboard also helps reveal hidden profitability patterns that may not be immediately visible from raw sales data alone.

An interactive dashboard is required because managers need to analyse business performance across multiple dimensions simultaneously, including regions, product categories, customer segments, discount levels and sales trends over time. First, the dashboard helps solve the problem of relying only on sales figures by allowing managers to compare both sales and profitability together. This makes it easier to identify issues such as profit leakage, excessive discounting and low-performing business areas. Second, the dashboard improves decision-making efficiency by using interactive filters and visualisations to present business insights more clearly than traditional static reports or tables. The dashboard supports business decisions related to discount management, product profitability, regional market evaluation and customer segment strategy, allowing managers to allocate resources more effectively and improve long-term business performance.

Page 3: Dataset Selection

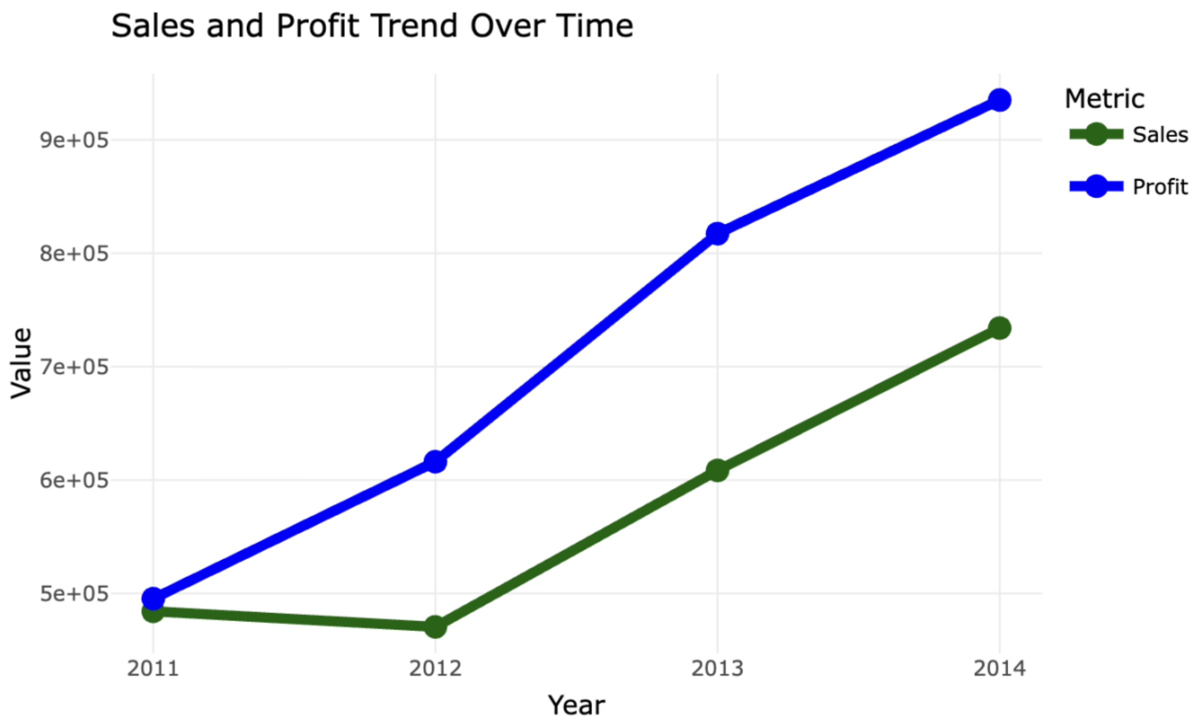
The main dataset used in this dashboard is the Superstore dataset. It contains transaction-level information including order date, sales, profit, discount, category, sub-category, customer segment, region and state. The dataset is suitable for analysing business performance because it allows sales and profitability to be compared across different products, customer groups and geographic areas.

The original datasets were processed before dashboard development to improve analytical relevance and data consistency. Data preparation included merging the external population dataset with the Superstore dataset, creating new business variables such as profit margin, discount bands and loss indicators, filtering U.S. records only, and removing unnecessary variables that were not required for dashboard analysis.

The external dataset used in this project is the U.S. state population dataset published by the U.S. Census Bureau (U.S. Census Bureau, 2026). The dataset contains annual resident population estimates for the United States, regions, states, the District of Columbia and Puerto Rico from 2020 to 2025. The main variables include state names and estimated resident population, with population measured in number of people. The dataset is publicly available through the U.S. Census Bureau Population Estimates Program and is widely used for demographic and regional analysis.

By integrating population data into the dashboard, regional performance can be interpreted more fairly rather than relying only on total sales. States with larger populations naturally have larger customer markets and may generate higher sales volumes even without stronger operational performance. Therefore, population-adjusted analysis such as sales per capita and profit per capita helps distinguish whether strong sales are driven by genuine market efficiency or simply by population size. This improves the dashboard's ability to support more effective regional evaluation, business expansion and long-term strategic decision-making.

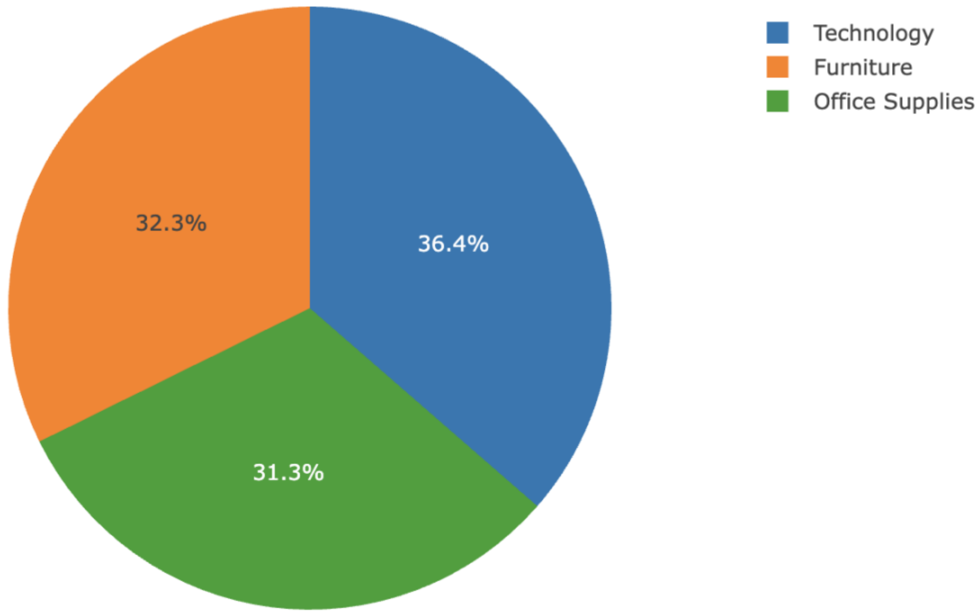
Page 4: Visualizations and Interpretation: Overall Sales and Profit Trend



The chart shows that both sales and profit increased from 2012 to 2014, indicating that Superstore experienced overall business growth during this period. Although sales declined slightly in 2012, both sales and profit rose strongly afterwards, especially between 2012 and 2014. This suggests that the company was not only generating higher revenue, but was also improving its ability to convert revenue into profit. Therefore, the visualization helps determine whether revenue growth is translating into actual profitability rather than simply reflecting increased sales volume.

This pattern is important because strong sales performance alone does not always indicate a healthy business. In many businesses, sales may continue to increase while profit remains weak due to excessive discounting, low-margin products, or inefficient operations. If sales grow without corresponding profit growth, the business may be sacrificing profitability in order to maintain revenue growth. However, in this dashboard, profit generally increased together with sales over time, suggesting that Superstore's growth during this period was relatively sustainable and financially healthier. This means the business was not only expanding revenue, but was also improving profitability performance at the same time. Therefore, managers should continue monitoring both sales and profit trends together, while carefully controlling discount strategies and low-margin products that could weaken future profitability.

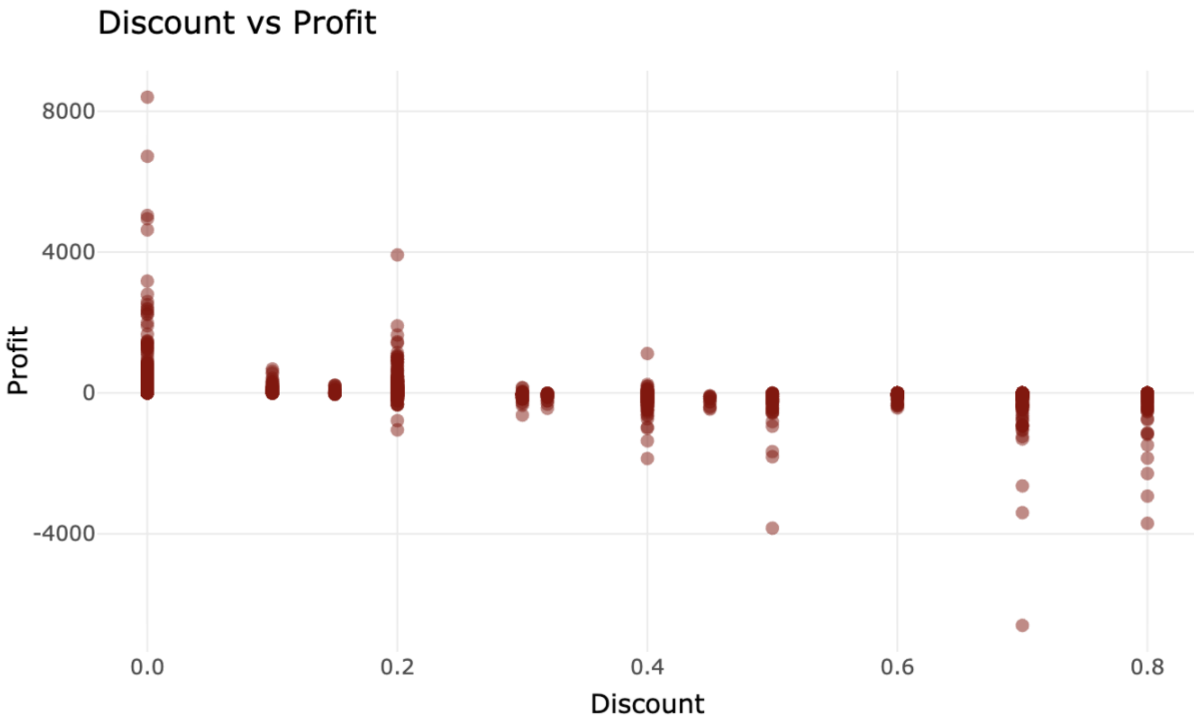
Page 5: Visualizations and Interpretation: Category Sales Contribution



The chart shows that Technology has the highest share of total sales among the three product categories, while Furniture and Office Supplies contribute slightly lower proportions. This indicates that Technology products are a major driver of Superstore's revenue performance. However, high sales alone should not be treated as strong business performance unless the category also contributes positive profit. Some product categories or sub-categories may generate strong sales volume but still produce weak profitability due to high costs, excessive discounting, or low profit margins.

This pattern is important because businesses can mistakenly focus only on revenue growth while ignoring whether products are actually generating sustainable profit. In Superstore, high sales in a category do not necessarily mean that the category is financially efficient. Certain sub-categories may still experience low profit or even losses despite contributing significant sales revenue. Therefore, managers should compare sales performance together with profit performance when evaluating product categories. The visualization suggests that managers should prioritise categories and sub-categories that achieve both strong sales and healthy profitability, while reviewing low-profit products, controlling discount strategies, and improving product mix decisions to strengthen long-term business performance.

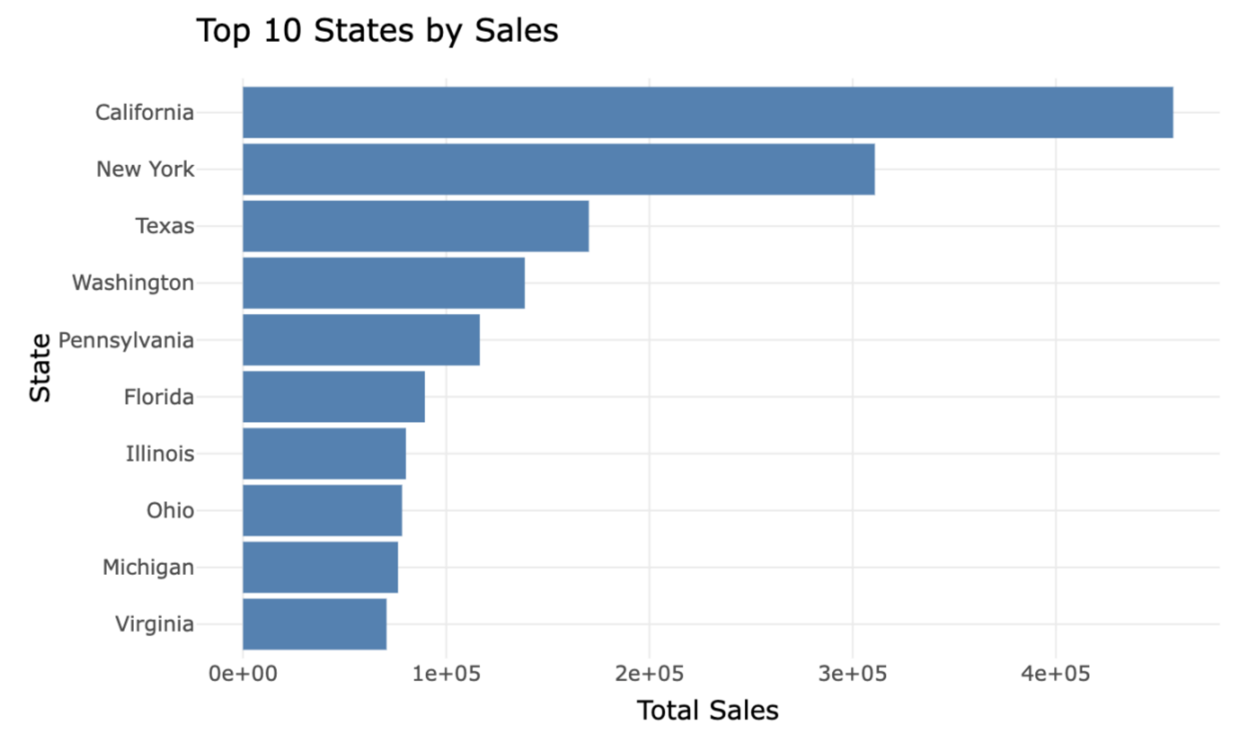
Page 6: Visualizations and Interpretation: Discount Impact on Profit



The chart shows a negative relationship between discount levels and profit performance. When discount levels are low, many transactions generate positive profit, and several orders achieve relatively high profitability. However, as discount levels increase, profit values become lower and more unstable, with many transactions moving into negative profit. This suggests that although discounting may help increase sales volume, excessive discounting can significantly reduce profitability and, in some cases, even lead to financial losses. Therefore, the chart suggests that discounting may be useful for generating sales, but excessive discounting can reduce or even eliminate profitability.

This pattern is important because businesses often use discounts to attract customers and increase revenue, but aggressive discount strategies may weaken long-term financial performance. The chart indicates that some products or categories are likely more sensitive to high discounting, especially products with already low profit margins. If managers focus only on increasing sales through discounts, the business may experience “high sales but low profit” problems. Therefore, managers should monitor discount levels carefully, identify categories or products that are heavily affected by discounting, and avoid applying excessive discounts to low-margin products. Instead, discounts should be used strategically to balance customer demand with sustainable profitability and overall business performance.

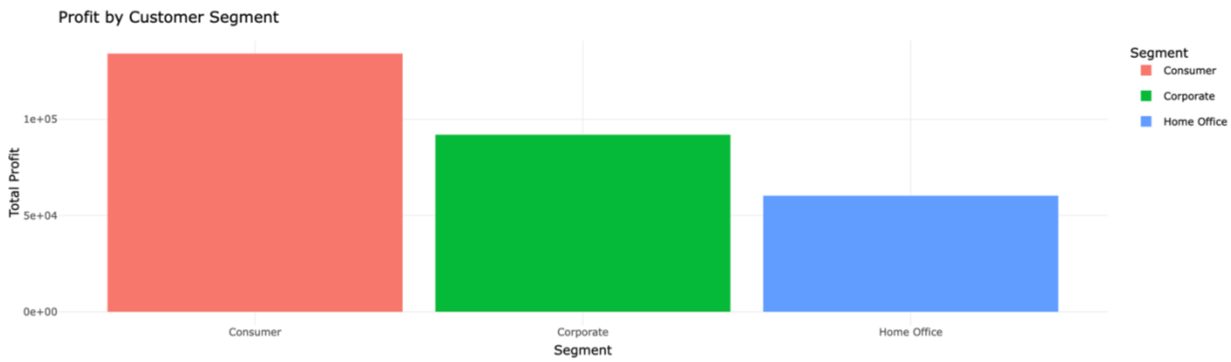
Page 7: Visualizations and Interpretation: Regional Performance Analysis



The chart shows that California, New York and Texas generated the highest total sales among all states, indicating that these regions are major contributors to Superstore's overall revenue performance. At first glance, these states may appear to be the strongest markets because they contribute significantly higher sales than other states. However, total sales alone may not provide a complete evaluation of regional performance because larger states naturally contain more consumers and larger market sizes.

After incorporating external population data from the U.S. Census Bureau, regional performance can be interpreted more fairly by comparing sales relative to state population size. Some highly populated states may achieve strong total sales mainly because of their large customer base, while smaller states with lower total sales may actually perform more efficiently on a per capita basis. Therefore, external population data changes the interpretation of regional performance by showing whether sales strength comes from genuine market efficiency or simply from population advantages. After adjusting for state population, regional performance can be evaluated more fairly because large states are no longer automatically treated as better-performing markets. This suggests that managers should evaluate both total sales and population-adjusted performance when making regional investment, expansion and marketing decisions.

Page 8: Visualizations and Interpretation: Customer Segment Profitability



The chart shows that the Consumer segment generates the highest total profit among all customer segments, followed by Corporate and Home Office. This indicates that Consumer customers contribute most strongly to Superstore's overall profitability and represent the company's largest source of business value. In contrast, the Home Office segment contributes the lowest total profit, suggesting that its overall profitability performance is comparatively weaker. However, total profit alone should not be used to evaluate customer segment performance because a segment with high revenue or large customer volume may still produce relatively weak profit margins.

This pattern is important because different customer segments may contribute differently to long-term business sustainability. Some segments may generate large sales volumes but require heavier discounting, higher service costs or lower-margin products, which can reduce profitability. Therefore, managers should compare customer volume together with profitability performance when evaluating segment strategy. The visualization suggests that Consumer customers currently provide the strongest profit contribution, while lower-performing segments may require more targeted pricing, marketing or product strategies to improve efficiency. This visualization helps managers identify whether customer volume or customer profitability should guide future strategy. As a result, managers can allocate resources more effectively and focus on customer segments that support stronger long-term profitability.

Pages 9-10: Data Story

The overall data story suggests that Superstore has experienced steady sales growth over time, indicating expanding customer demand and market activity. However, stronger sales performance does not automatically guarantee stronger profitability. Although both sales and profit increased in the later years, the dashboard shows that business performance should not be evaluated based on revenue alone, because profitability may still be affected by discounting strategies, product mix and operational efficiency across different areas of the business. This means that revenue growth does not always translate into equally strong financial performance. Therefore, the dashboard highlights the importance of analysing both sales and profit together in order to better understand the overall health and sustainability of the business.

The product category and discount analysis suggests that strong sales performance does not always lead to strong profitability. The dashboard shows that some product categories generate high sales contributions, but profitability may still remain limited when discount levels become too high. This pattern is commonly observed in the retail industry, where businesses often rely on aggressive discount strategies to increase customer demand and short-term sales volume. For example, J.C. Penney experienced significant profitability challenges due to its long-term reliance on discount-driven retail strategies, which weakened its pricing power and customer perception (Panos Mourdoukoutas, 2017). Similarly, the relationship between discount and profit in the dashboard indicates that higher discount levels are generally associated with lower or even negative profitability outcomes. This suggests that Superstore may be experiencing profitability leakage, where revenue growth is partially achieved at the expense of profit stability. Therefore, management should not evaluate performance based on sales volume alone, but also consider whether pricing and discount strategies are generating sustainable profitability. Greater attention should be given to controlling excessive discounting and prioritising products that contribute stronger long-term profit performance.

The regional analysis becomes more meaningful after incorporating the external population dataset because total sales alone may not accurately represent market performance. The dashboard shows that states such as California and New York generate the highest sales volumes, but these states also have some of the largest populations in the United States. In real business environments, highly populated markets naturally generate stronger sales because they contain larger customer bases and stronger purchasing demand. For example, higher sales performance among major retailers such as Walmart and Target is often associated with larger population markets and economies of scale in densely populated states, rather than purely superior operational performance (U.S. Census Bureau, 2026; Miao, 2023). Therefore, relying only on total sales may create misleading conclusions about which markets are truly performing efficiently. By integrating population data into the dashboard, the analysis can better distinguish whether strong sales are driven by genuine market efficiency or simply by population advantages. This means that some smaller states with lower total sales may still demonstrate relatively strong performance when adjusted for population size. As a result, managers should evaluate both market size and profitability efficiency when making regional investment and expansion decisions in order to identify markets with stronger long-term business potential.

The customer segment analysis suggests that different customer groups contribute differently to Superstore's profitability performance. The dashboard shows that the Consumer segment generates the highest total profit, while Corporate and Home Office contribute relatively lower profit levels. However, higher customer volume or revenue does not always indicate stronger profitability because some customer segments may require heavier discounting, lower pricing strategies or higher service costs. This pattern is also common in business environments, where some customer segments contribute strongly to sales activity but may generate weaker long-term profitability because of higher operational or promotional expenses (Anokhin et al., 2026). Therefore, businesses should not evaluate customer segments based only on sales or customer size alone. Instead, managers should focus on identifying customer groups that consistently contribute stronger profitability and sustainable business value. This visualization helps managers determine whether future strategy should prioritise customer volume or customer profitability, allowing resources and marketing efforts to be allocated more effectively toward higher-value customer segments.

Overall, the data story suggests that Superstore should focus on achieving profitable growth rather than simply increasing sales volume. Although the business demonstrates strong sales performance across different regions, product categories and customer segments, the dashboard indicates that profitability is strongly influenced by discount strategy, product mix and regional market efficiency. Therefore, management should focus not only on revenue expansion, but also on improving the sustainability of profit generation. Excessive discounting should be controlled more carefully, while greater attention should be given to products and customer segments that consistently generate stronger profitability. In addition, the integration of external population data highlights the importance of evaluating regional performance more fairly rather than relying only on total sales figures. Overall, the dashboard demonstrates that long-term business success depends on balancing revenue growth, profitability and market efficiency simultaneously.

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